AGROSHIFT GAMBIA

Farmer Survey Analysis & Platform Technical Recommendations

Preliminary analysis of the submitted farmer questionnaire dataset | 6–7 November 2026

Prepared as a research-to-product document for the AgroShift Gambia research and technical teams.

Executive summary

The submitted file contains 3 completed farmer responses across 49 exported columns. All three respondents consented. The dataset provides useful early discovery evidence, but the sample is very small and heterogeneous: it includes smallholder vegetable production, drip-irrigated vegetable production and a fresh-produce export operation. Findings must therefore be treated as exploratory signals, not estimates of the wider farming population in The Gambia.

Three product implications stand out. First, market demand is central to crop choice for all three respondents. Second, the respondents report environmental and production constraints, including heat, declining fertility, pests/diseases, water constraints, input costs and labour. Third, a single generic recommendation pathway will not serve all users: AgroShift should support at least smallholder/local-market and commercial/export profiles.

Recommended technical direction: build a transparent, rule-based crop-sequence recommendation engine using hard agronomic constraints followed by a weighted multi-criteria ranking (MCDA). Do not begin with a machine-learning model trained on this three-row dataset. Recommendations should show their reasons, data sources, assumptions, confidence and any missing information.

1. Dataset overview and scope

Item	Observed	Notes
Rows / respondents	3	Three valid submitted records were present in the CSV.
Exported columns	49	Includes timestamp, consent and questionnaire items; Q42 is absent from the exported question numbering.
Consent	3/3 yes	All respondents agreed to participate.
Collection dates	6–7 Nov 2026	Based on timestamps in the file.
Geographic coverage	Niumi Sittanunku (NBR), Gunjur, Nemakungu/Sukuta Area	Only three named locations; not geographically representative.
Farming systems	Irrigated, drip irrigation, irrigated export operation	The sample does not currently represent a broad rain-fed farmer segment.

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2. Descriptive findings

2.1 Main patterns

Market demand: 3 of 3 respondents (100%) rated market demand as very important when choosing crops. This suggests that market feasibility should be a core filter, not an optional extra.

Weather change: 3 of 3 reported that weather changes affect farming. Reported changes include higher temperatures; one respondent also listed delayed/early rainfall patterns, dry periods, intense rainfall, flooding, drought, wind, pests and diseases.

Soil fertility concern: 3 of 3 selected soil becoming less fertile. Self-rated fertility ranged from very fertile to moderately fertile, showing that perceived current fertility and perceived trend are different variables.

Crop failure: 2 of 3 answered yes to major crop failure in the last three seasons. Reported causes included late planting and lack of labour. One respondent answered no but still supplied 'Disease' in the conditional cause field; this needs data cleaning and a form-logic fix.

Seed access: All three described a seed-supply issue: very difficult, difficult, or all seeds imported. Seed availability should be checked before a new crop is recommended.

Rotation motivations: Respondents mentioned reducing disease, improving production, soil condition and yield. Reported benefits included better crop growth, higher income and better soil condition.

Preferred information: One prefers WhatsApp, one a smartphone application, and one a government advisory channel. This supports multi-channel delivery rather than app-only delivery.

Recommendation explanation: Two respondents explicitly want an explanation for recommendations; one does not. Explanations should be available by default but optionally collapsible.

2.2 Differences between the respondent profiles

Respondent F01 (Niumi Sittanunku) is a small-scale irrigated vegetable farmer reporting pest/disease, heat, fertiliser/seed cost, equipment and water constraints. Their stated field size is '200m square' with square metres selected; confirm whether this means 200 m^2 or a 200 m × 200 m plot.

Respondent 001 (Gunjur) reports drip irrigation, a borehole, vegetable production and a field size entered as '150/250' while selecting hectares. This is ambiguous and likely requires direct confirmation before any area-based calculation is used.

Respondent Nema1 (Nemakungu/Sukuta) represents a commercial fresh-produce export operation reporting 35 hectares, export-only sales, imported seed, buyer-led crop choice, and crops including peppers, babycorn, French beans and mangoes. This respondent explicitly notes that a locally oriented questionnaire does not fully fit export farms.

3. Data quality issues to resolve before building analytics

Issue	Recommended action
Very small sample (n=3)	Do not publish population percentages or claim national farmer preferences. Label all charts 'among 3 respondents'. Continue sampling across regions, farm types and rain-fed/irrigated systems.

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Field area ambiguity	F01: '200m square' needs confirmation. Farmer 001: '150/250' with hectares needs clarification. Store numeric area and unit separately; calculate hectares only after confirmation.
Missing farmer ID	First respondent has no farmer identification code. Use a generated pseudonymous ID (e.g., F01) in the analysis dataset; do not expose names or contact details.
Experience field formatting	One answer is exported as 'Option 1Less than 1 year'. Standardise to a controlled value and correct the Google Form option label.
Crop history too broad	Answers such as 'Vegetables' are not enough to calculate rotation or crop-family breaks. Collect individual crop names, season/year, plot ID and sequence.
Contradictory rotation answer	Exporter answered 'No, I usually grow the same crop' but described same crops rotating to improve soil and yield. Clarify whether rotation means changing crop species on the same plot, changing crop varieties, or alternating with fallow.
Conditional question mismatch	A respondent who answered 'No' to crop failure still supplied a cause. Make the cause question conditional on 'Yes' and treat this existing answer as unresolved, not as a confirmed failure.
Exporter segment mismatch	Add a dedicated commercial/export module covering buyer contracts, accepted crops/varieties, grades, residue/food-safety requirements, planting windows, volume commitments, cold chain and price/market risk.
Missing plot coordinates and soil tests	Village-level location is not sufficient for plot-level environmental extraction. Collect consented GPS coordinates or polygon, coordinate accuracy, soil sample results and sampling date.
Question numbering gap	The exported questions jump from Q41 to Q43. Check whether Q42 was intentionally removed or omitted.

4. Product requirements derived from the survey

Feature/data domain	Minimum requirement
Farmer and farm profile	Capture farmer type (smallholder/commercial/export), community, farming system, plot identifier, farm experience, farm area and unit.
Plot-level location	Capture latitude/longitude or field boundary polygon with informed consent, GPS accuracy, capture date and source. Allow manual location selection if GPS is unavailable.
Crop history	Capture each crop by season/year and plot, including planting date, harvest date, variety, crop family, yield, area and whether the plot was fallow/cover-cropped.
Soil and land	Capture soil texture, pH, organic carbon/organic matter, drainage, salinity where relevant, erosion, compaction, fertility observations and lab/sample provenance.
Water and climate	Capture water source, irrigation type, reliability, water restrictions, rainfall/temperature history and seasonal forecasts where available.

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Inputs and resources	Capture seed/variety availability, fertiliser/manure/compost, pesticide/herbicide use, labour availability, machinery/equipment and indicative costs.
Market and economics	Capture local buyer demand or export buyer requirements, expected farm-gate price, costs, yield assumptions, market access, contract/volume constraints and price uncertainty.
Recommendation preferences	Capture farmer objectives and weights: yield, net income, soil health, cost, water use, pest/disease risk, labour and market reliability.
Recommendation explanation	Show why a sequence is suggested, key trade-offs, required inputs, assumptions, evidence level and what the farmer should confirm with an extension officer.
Feedback and monitoring	Allow farmer/extension officer to record whether the recommendation was accepted, planted, changed, harvested and what yield/cost/outcome resulted.

5. Recommended crop-rotation decision method

Use a hybrid decision-support approach: (A) rule-based agronomic screening to reject unsafe or infeasible sequences, followed by (B) weighted Multi-Criteria Decision Analysis (MCDA) to rank the feasible alternatives. This is transparent, explainable and appropriate while the project is still building a verified local knowledge base.

5.1 Step A – hard constraints / eligibility filters

- Crop must fit the farmer's planting window and local growing season.
- Crop must be compatible with the plot's water availability and irrigation capacity.
- Crop must be suitable for the measured/estimated soil conditions and drainage; do not treat coarse map data as a soil test.
- Crop family and disease/pest history must be checked against the previous crop sequence.
- Required seed/variety must be obtainable in time and suitable for the market.
- Labour, equipment, input and budget requirements must be feasible.
- For export mode, crop/variety must meet buyer demand, quality, certification and delivery requirements.

5.2 Step B – rank eligible sequences using MCDA

For each candidate sequence, score each criterion from 0 (poor) to 5 (excellent), using documented agronomic rules and local data. Example starting weights below are illustrative only and must be validated with farmers and agronomists.

Illustrative criterion	Initial weight
Agronomic suitability / soil fit	25%
Market fit / buyer acceptance	20%
Water and climate fit	15%
Pest and disease break / crop-family diversity	15%
Expected net return	10%
Seed, labour and input feasibility	10%
Farmer preference / adoption fit	5%

Weighted score = Σ (criterion score ÷ 5 × criterion weight). Total score is expressed out of 100. Keep hard constraints separate: a high weighted score must never override a failed safety/feasibility constraint.

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5.3 Smallholder and export profiles

Smallholder profile: priorities crop suitability, soil improvement, seasonal risk, seed access, affordability, household/labour fit and local market demand. Export/commercial profile: make buyer acceptance and crop/variety specification mandatory filters, then consider production schedule, volume, quality, labour, water, input cost, export logistics and expected margin. Allow the user or adviser to adjust weights transparently.

5.4 Recommendation output

- Top 3 feasible crop sequences, not a single unexplained answer.
- For each sequence: crop order, season/calendar, crop family, planting/harvest window and fallow/cover-crop period where appropriate.
- Why it is recommended: soil, water, climate, pest/disease and market reasoning.
- Trade-offs: costs, labour, seed availability, water demand, market uncertainty and possible yield/income ranges.
- Confidence label and missing-data warning (e.g., high/medium/low evidence; never imply statistical certainty without validation).
- Citations/knowledge-base references and date of last review; provide an extension-officer review pathway.

6. Data fields the technical team should priorities

Entity	Suggested fields
Farmer	farmer_id, consent_status, farmer_segment, preferred_language, contact_channel (optional)
Farm/plot	farm_id, plot_id, community, latitude, longitude or polygon, GPS_accuracy_m, area_value, area_unit, land_type, irrigation_type
Season/crop history	season_id, plot_id, crop_id, variety, crop_family, planting_date, harvest_date, area_planted_ha, yield_value, yield_unit, fallow/cover_crop
Soil	sample_id, plot_id, sample_date, pH, organic_carbon, texture, N/P/K if tested, salinity/EC if relevant, lab/provider, method
Climate/water	date, rainfall_mm, tmin_c, tmax_c, water_source, irrigation_reliability, water_limit
Resources/economics	seed_available, seed_cost, fertiliser_cost, labour_days, labour_cost, equipment_access, expected_price, buyer_acceptance
Recommendation	recommendation_id, rule_version, candidate_sequence, criteria_scores, weights, total_score, exclusions, explanation, confidence, created_at
Outcome	planted_sequence, actual_dates, harvested_yield, realised_price, input_cost, pest/disease observations, user_feedback

7. Revised questionnaire recommendations

- Keep a common core questionnaire for all farmers, then route respondents to smallholder, commercial/export, livestock-integrated or irrigated/rain-fed modules as relevant.
- Replace free-text field area with a numeric value plus unit dropdown; add a simple example and a validation range.

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- Ask crop history as a repeated structured record: crop, season/year, plot, planting date, harvest date, variety and yield. Do not rely only on a multi-crop text box.
- Ask separately whether the farmer changes crop species on the same plot, changes variety, interplants/mixes crops, or uses fallow/cover crops.
- Make crop-failure cause conditional on a 'Yes' response; allow multiple causes and 'not sure'.
- Collect GPS point/polygon only with explicit consent and explain how location will be used.
- Add soil-test availability and date; distinguish measured soil data from farmer perception.
- Add market questions for local sales and a separate export buyer/specification module.
- Add expected costs, selling prices and yields as ranges if exact figures are sensitive.
- Add a feedback section to record whether farmers accept, try and benefit from a recommendation.

8. Immediate decisions for the project team

1. Confirm whether the MVP targets smallholder farmers only or both smallholder and commercial/export farms.
2. Agree that recommendations are plot-specific and season-specific, not just village-level.
3. Decide which crops and regions are in the first validated crop knowledge base.
4. Identify an agronomist/extension officer to approve rotation rules and crop calendars.
5. Collect plot coordinates, field area confirmation and structured three-season crop histories from pilot farmers.
6. Adopt a transparent rule-based + MCDA engine for the first release; postpone ML until data supports it.
7. Define a safe fallback: when key soil, water, market or crop-history data is missing, show a provisional advisory or request expert review rather than issuing a confident rotation.

9. Conclusion

The survey has already revealed a key design principle: a useful crop-rotation tool must balance agronomic suitability with real-world feasibility. Soil health matters, but farmers also need seeds, labour, water, affordable inputs and a reliable market. AgroShift should therefore operate as an explainable decision-support system, not an automated crop selector. The immediate priority is to improve structured farm/plot data, establish a locally validated crop knowledge base, and test a transparent rules-plus-MCDA prototype with farmers and extension officers.

References and technical links

https://docs.google.com/forms/d/e/1FAIpQLSdow4wyVe-P_IYhD9dEXf5efjHedUDhCRoYrAFOYpmhiJr8sg/viewform?usp=dialog

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